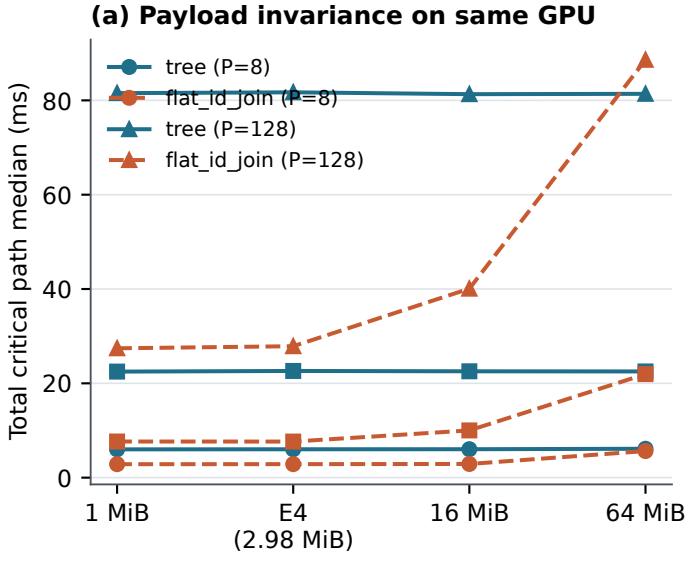
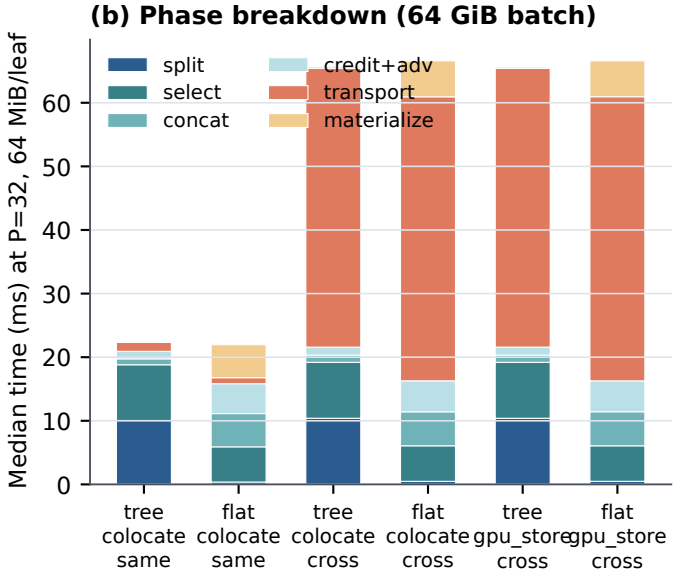
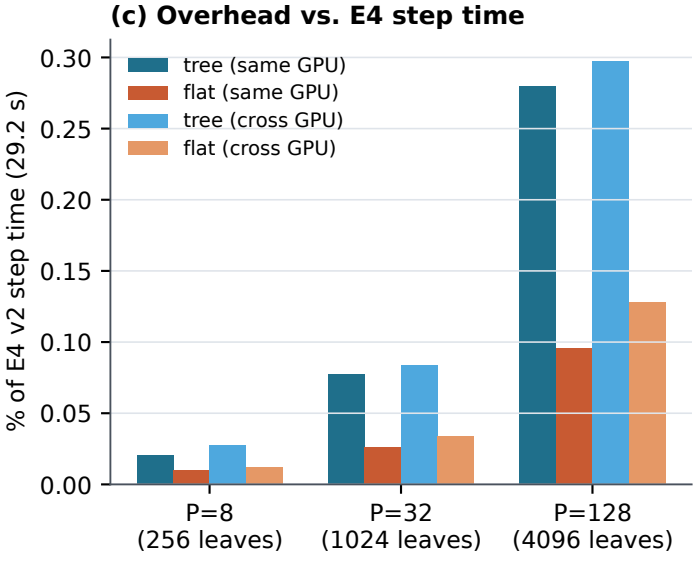

(a) Payload invariance on same GPU
Total critical path median (ms)
tree (P=8)
flat_id_join (P=8)
tree (P=128)
flat_id_join (P=128)
0
20
40
60
80
1 MiB
E4
(2.98 MiB)
16 MiB
64 MiB


(b) Phase breakdown (64 GiB batch)
Median time (ms) at P=32, 64 MiB/leaf
split
select
concat
credit+adv
transport
materialize
0
10
20
30
40
50
60
tree colocate same
flat colocate same
tree colocate cross
flat colocate cross
tree gpu_store cross
flat gpu_store cross


(c) Overhead vs. E4 step time
% of E4 v2 step time (29.2 s)
tree (same GPU)
flat (same GPU)
tree (cross GPU)
flat (cross GPU)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
P=8
(256 leaves)
P=32
(1024 leaves)
P=128
(4096 leaves)